# Hazard Analysis
# Squigglr

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Table 1: Revision History

| **Date** | **Developer(s)** | **Change** |
|---|---|---|
| 14 October 2026 | Rayan Nasrallah | Added critical assumptions CA-001 to CA-006 and hazards H-RAY-001 to H-RAY-007 (#91). |
| 15 October 2026 | Rayan Nasrallah | Added my reflection responses to the appendix and my generative AI use disclosure (#140). |
| 15 October 2026 | Ahmed Elzaria | Added drawing-input and guided-lesson FMEA hazards H-AHM-001 to H-AHM-007 (part of #135). |
| 16 October 2026 | Moustafa Moustafa | Added sensory, feedback, progress-view and refinement hazards H-MOU-001 to H-MOU-006 (#115). |
| 17 October 2026 | Moustafa Moustafa | Added my reflection responses to the appendix and my generative AI use disclosure (#117). |

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[You are free to modify this template. —SS]

# 1 Introduction

[You can include your definition of what a hazard is here. —SS]

# 2 Scope and Purpose of Hazard Analysis

[You should say what **loss** could be incurred because of the hazards. —SS]

# 3 System Boundaries and Components

[Dividing the system into components will help you brainstorm the hazards. You shouldn't do a full design of the components, just get a feel for the major ones. For projects that involve hardware, the components will typically include each individual piece of hardware. If your software will have a database, or an important library, these are also potential components. —SS]

# 4 Critical Assumptions

The assumptions below are kept few and are written so that they do not remove hazards. Squigglr does not assume that an adult watches the child during practice, that the network is always available, or that the device meets Squigglr's requirements; those conditions are treated as failure modes in the Failure Mode and Effect Analysis (FMEA). Sources are cited by their identifiers in the Software Requirements Specification (SRS), Appendix A, and interview turns by speaker and timestamp, for example (SRC-005, P1, 34:13). Each assumption lists the H-RAY hazards it affects.

**CA-001** An adult sets up Squigglr for a child, chooses and starts the lesson and its starting stage, and is the only person meant to accept a suggestion to go back a stage. Within a lesson the child does not navigate, exit or restart; Squigglr moves between stages. An adult may not be present or watching while the child practises.
*Rationale:* children should practise without constant adult help, while adults handle setup (SRC-005, P1, 34:13), and adults decide progression (SRC-005, P1, 28:37; CON-007).
*Hazards affected:* H-RAY-001, H-RAY-004, H-RAY-005.

**CA-002** The device, browser, touchscreen or stylus in use may not be one of the supported combinations (`SUPPORTED_DEVICES`, TBD), and Squigglr may be opened on a device without a stylus, where the child can draw with a finger but not as intended (ASM-003).

*Rationale:* Squigglr is a website opened from a web address with no installation (CON-001, CON-002), so the team does not control the device. The supported combinations are not yet defined (CON-003), and the team did not promise that Squigglr works on every device (SRC-005, Moustafa, 6:03).
*Hazards affected:* H-RAY-002.

**CA-003** The network connection may be slow or lost at any time, including during an attempt.
*Rationale:* attempts are saved to an account and reviewed later from another device, and offline access was not promised (SRC-005, Moustafa, 6:03). Home and school networks are outside the team's control.
*Hazards affected:* H-RAY-002, H-RAY-006.

**CA-004** One device may be used by several children, at school or at home.
*Rationale:* school devices may be shared by several children (SRC-004), and Squigglr may be used at home, at school or in other settings such as rehabilitation centres (SRC-005, P2, 21:25).
*Hazards affected:* H-RAY-006.

**CA-005** The adults who read practice records are not assumed to have training in visual-motor development, and they may show the records to other people, such as teachers or therapists.
*Rationale:* parents and caregivers are not assumed to be specialists (SRS, Hands-On Users), and records exist to support conversations about what a child needs (SRC-005, Moustafa, 6:03). Squigglr makes no diagnosis or treatment claims (SRC-001).
*Hazards affected:* H-RAY-003.

**CA-006** The supporting adult chooses the adult-set values, such as `STAGE_PASS`, `DIFFICULTY_STREAK` and `ATTEMPT_LIMIT`, for each child within a range the team sets (SRC-008). The ranges are not found from child testing, and the value an adult picks, or the default, may not suit their child.
*Rationale:* no information is gathered from children and no study with children is run (CON-005, CON-006), so the ranges cannot be tuned on children (SRS, Goals of the Project). Children reach these skills at different points (SRC-005, P1, 20:27).
*Hazards affected:* H-RAY-004, H-RAY-005, H-RAY-007.

# 5 Failure Mode and Effect Analysis

[Include your FMEA table here. This is the most important part of this document. —SS] [The safety requirements in the table do not have to have the prefix SR. The most important thing is to show traceability to your SRS. You might trace to requirements you have already written, or you might need to add new requirements. —SS] [If no safety requirement can be devised, other mitigation

strategies can be entered in the table, including strategies involving providing additional documentation, and/or test cases. —SS]

## 5.1 Supervision, Context of Use and Use of Records (H-RAY)

Table 2 covers hazards from how Squigglr is supervised, where and on what it is used, and how its records are read. Each recommended action starts with Eliminate, Mitigate or Document/Test. Mitigating requirements trace to the goals (G.#) and constraints (CON-###) in the SRS, or to safety requirements SAF-RAY-### in the SRS, Safety-Critical Requirements.

Table 2: FMEA: supervision, context of use and use of records

| **Hazard ID** | **Component** | **Failure mode** | **Cause(s)** | **Effect and loss** | **Category** | **Recommended action** | **Mitigating requirement IDs** |
|---|---|---|---|---|---|---|---|
| H-RAY-001 | Lesson flow; adult controls | A child practises with no adult watching, or the adult is inattentive, and the child reaches an adult-only action. | Children are meant to practise without constant adult help (CA-001); adult controls use the same touch interface as the child. | The child accepts a suggestion to go back a stage, or changes lesson or stage, without an adult's decision; the child's distress goes unnoticed. Loss: progression not chosen by an adult; well-being. | Psychological/ well-being | Mitigate: place every adult-only action (choosing a lesson or stage, accepting a suggestion) behind an adult check that a child cannot pass by tapping icons; keep a pending suggestion waiting, without blocking practice, until an adult reviews it. | CON-007, G.3, SAF-RAY-001 |
| H-RAY-002 | Browser front end; saving of attempts | Squigglr is used on an unsupported device or browser, without a stylus, or loses its connection during an attempt, at home, at school or on a shared device. | Device not controlled by the team (CA-002); no offline access (CA-003). | Strokes are not captured correctly, the child's attempt is lost, or a lesson cannot be completed; the child is frustrated and the record is incomplete. Loss: lost practice data; availability. | Data integrity or Availability | Mitigate: before a lesson starts, check the device and browser against `SUPPORTED_DEVICES` and show the adult a notice when they do not match; keep each attempt until the server confirms it is saved, and tell the adult when an attempt could not be saved. | CON-001, CON-002, CON-003, G.5, SAF-RAY-002, SAF-RAY-003 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-RAY-003 | Attempt records; progress summaries | Practice records are used as clinical or diagnostic evidence, or as a measure of the child's ability. | Adults reading records lack specialist training, and records may be passed to others (CA-005); fewer attempts may be read as better skill (SRC-005, Moustafa, 6:03). | An adult or professional draws a wrong conclusion about the child, such as a label or a placement decision. Loss: harm to the child's well-being and to decisions about support. | Psychological/ well-being | Eliminate: show no diagnostic labels, judgements of the child's abilities or comparisons with other children in any record or summary. Mitigate: state on every summary and export that the records describe practice only and are not an assessment. | G.5, G.6, SAF-RAY-004 |
| H-RAY-004 | Lesson flow; feedback | Repeated failed attempts at one stage continue until the child becomes frustrated or agitated. | Failure feedback repeats; `ATTEMPT_LIMIT` may be too high for a given child (CA-006); an improvement extension lengthens the lesson; no adult may be watching (CA-001). | The child becomes distressed, stops practising or refuses the activity later. Loss: well-being; the practice benefit. | Psychological/ well-being | Mitigate: end the lesson at `ATTEMPT_LIMIT`, or at `ATTEMPT_LIMIT_MAX` when extended, and save progress (G.3); let audio feedback be muted (SRC-005, P2, 37:39). Document/Test: test that each lesson ends at `ATTEMPT_LIMIT` and that no extended lesson passes `ATTEMPT_LIMIT_MAX`. | G.3, SAF-RAY-005 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-RAY-005 | Lesson flow; attempt records | After a lesson ends at `ATTEMPT_LIMIT`, it is restarted at once and the repetition continues. | No adult may be watching (CA-001); a restart starts a new count of attempts. | The limit does not stop the repetition, and the adult cannot see how often it happened. Loss: well-being; misleading records. | Psychological/ well-being | Mitigate: record each lesson that ended at `ATTEMPT_LIMIT` and show the adult how many times this happened per lesson; keep earlier attempts when a lesson is restarted. Document/Test: test that a restart after the limit keeps the earlier record. | G.3, G.5, SAF-RAY-006 |
| H-RAY-006 | Learner selection; attempt records | On a shared device, attempts or a pending suggestion are saved to, or shown for, the wrong child. | Several children use one device (CA-004); the active child is not changed between sessions; a lost connection leaves a session open (CA-003). | One child's records are shown to another child or adult, or a child's history is mixed with another's, so an adult accepts a suggestion based on the wrong attempts. Loss: privacy; record integrity. | Privacy/ security | Mitigate: show which learner is active on every practice screen in a way a child who cannot read can recognize; require an adult action to switch learners; show the attempts behind a suggestion when the adult reviews it. | G.5, SAF-RAY-001, SAF-RAY-007 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-RAY-007 | Child profile settings; lesson flow | A stage check or lesson limit uses a value that does not suit the child: `STAGE_PASS` too strict or too lenient, or `ATTEMPT_LIMIT` not larger than `DIFFICULTY_STREAK`, or `ATTEMPT_LIMIT_MAX` below `ATTEMPT_LIMIT`. | The adult misjudges the example drawings or leaves a default that does not suit the child (CA-006); a value is saved outside its range; a child changes a setting. | A child cannot pass a stage they can draw, or passes one they cannot, and the records mislead the adult. Loss: well-being; misleading records. | Psychological/ well-being | Mitigate: accept only values within the team-set range, and only an `ATTEMPT_LIMIT` larger than `DIFFICULTY_STREAK`; show example drawings of each end of the `STAGE_PASS` range; place settings behind the adult check; show the values in use next to the child's records. Document/Test: test that out-of-range and invalid values are refused and that the default is the midpoint. | G.2, G.3, SAF-RAY-008 |

## 5.2 Drawing Input and Guided-Lesson Decisions (H-AHM)

Table 3 examines failures in the capture and interpretation of stylus input and in the decisions that use it. These failures can occur even on an otherwise supported device with a working network, so they are distinct from the environmental failure in H-RAY-002. Effects are assessed against drawing accuracy, record integrity and the child's well-being; Squigglr does not make clinical assessments (BR-006). Recommended actions are traced to the learning-activity requirements in the SRS. Two additional safeguards, SAF-AHM-001 and SAF-AHM-002, are specified in the SRS Safety-Critical Requirements section.

Table 3: FMEA: drawing input and guided-lesson decisions

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-AHM-001 | Stylus input capture; attempt state | A stroke or part of one stroke disappears during a multi-stroke attempt, but the attempt is still finalized and evaluated. | Missed pointer samples, a brief input interruption or premature finalization (ASM-004); a stylus lift is mistaken for the end of an attempt. | The drawing appears incomplete or is scored inaccurately; a child receives undeserved failure feedback or loses a practice attempt. Loss: accurate feedback and record integrity. | Data integrity | Mitigate: retain ordered points and stroke boundaries; distinguish stylus lifts from finalization; treat interrupted or unevaluable input as non-passing without consuming an attempt. Document/Test: replay partial strokes and interruptions before and during a stroke. | FR-LRN-007, FR-LRN-008, FR-LRN-009, FR-LRN-010 |
| H-AHM-002 | Stylus event capture; attempt evaluation | One drawn stroke is captured or evaluated twice, inflating coverage or producing a spurious extra line. | Duplicate input events, accidental replay after reconnection, or combining the same stroke twice. | A poor attempt may pass, or a correct one may fail; saved strokes no longer represent what was drawn. Loss: integrity of feedback and practice records. | Data integrity | Mitigate: reject duplicate events within the same attempt before evaluation, without deleting legitimate separate strokes. Document/Test: replay duplicate events and distinct repeated strokes; verify that each is handled appropriately. | FR-LRN-007, FR-LRN-008; SAF-AHM-001 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-AHM-003 | Input coordinates; reference-shape alignment | Valid stylus points are mapped to the wrong position or order and used to score the drawing. | Wrong canvas scale, coordinate offset or incorrect ordering after a resize or orientation change (ASM-004). | An accurate attempt fails, an inaccurate attempt passes, or incorrect strokes are saved. Loss: data integrity and trustworthy progress records. | Data integrity | Mitigate: verify that captured coordinates remain in the current drawing surface and retain their recorded order; reject input that cannot be mapped reliably. Document/Test: repeat known stroke traces before and after resizing or rotating the device. | FR-LRN-007, FR-LRN-010; SAF-AHM-002 |
| H-AHM-004 | Completeness and pass classifier; lesson flow | An incomplete drawing (for example, a square missing a side) is reported as passing and advances the child. | The geometric result is high even though a required segment is missing, or completeness is not checked before threshold comparison. | The system advances despite an unfinished task and records a misleading success. Loss: record integrity and appropriate guidance. | Data integrity | Eliminate: require completeness and the adult-selected passing threshold to both hold before a pass; otherwise remain at the same stage. Document/Test: use complete and missing-part drawings immediately above and below the threshold, including a high-scoring incomplete example. | FR-LRN-011, FR-LRN-013, FR-LRN-023 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-AHM-005 | Feedback-selection logic; stage-result mapping | A detected attempt result or drawing difficulty is associated with the wrong pre-approved feedback observation (for example, a missed endpoint is reported as a missing segment). | A classification identifier is mapped to the wrong feedback entry, or the feedback lookup uses the wrong lesson or guidance stage. | The child is directed to correct a part of the drawing that was not identified as incorrect; the reported observation does not match the evaluated attempt. Loss: accurate guidance and feedback integrity. | Data integrity | Mitigate: select feedback from the approved catalogue using the evaluated result and correct lesson and stage identifiers; when no matching observation exists, do not present an unrelated correction. Document/Test: check the mapped message for every predefined classification and each drawing stage; test missing and mismatched identifiers. | FR-LRN-011, FR-LRN-013, FR-LRN-014 |
| H-AHM-006 | Failure-streak and suggestion logic; adult decision | Squigglr recommends moving backward at the wrong time, or changes the stage without the adult accepting the suggestion. | Invalid input is counted as failure, the streak is not reset, or the improvement suppression rule is omitted (BR-003). | The child receives unsuitable support or a needless interruption; adult control is bypassed. Loss: well-being and authorized lesson progression. | Psychological/ well-being | Mitigate: base the suggestion only on counted non-passing attempts and the defined improvement predicate; leave the current stage unchanged until an authorized adult accepts. Document/Test: replay improving, non-improving, invalid and threshold-boundary sequences; decline or ignore a suggestion. | FR-LRN-009, FR-LRN-018, FR-LRN-019, FR-LRN-020 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-AHM-007 | Learner settings; attempt and lesson evaluation | The active attempt is evaluated using another learner's threshold, a stale value or a setting different from the adult's saved selection. | Wrong profile linkage, settings not refreshed after an adult change, or defaults applied over saved choices (BR-008; CA-006). | A drawing passes or fails incorrectly, a lesson ends too early or continues too long, and the adult receives misleading records. Loss: well-being and record integrity. | Data integrity | Mitigate: use the active learner's validated saved settings for passing, suggestions and limits; use midpoint defaults only for unset values. Document/Test: switch learner profiles and change settings; test boundaries, invalid combinations and the exact saved-versus-default choice. | FR-LRN-016, FR-LRN-017, FR-LRN-019, FR-LRN-021 |

### 5.3 Sensory Load, Feedback, Progress Views and Refinement (H-MOU)

Table 4 covers hazards from how Squigglr looks, sounds and speaks to the child, how its progress views describe the child to adults, and how it changes the child's creative drawings. These hazards arise even when input, saving and lesson logic work correctly. H-AHM-005 covers feedback mapped to the wrong observation, H-RAY-003 records read as a diagnosis, and H-RAY-004 frustration from repeated failure; the rows below do not repeat them. Mitigating requirements trace to the usability requirements (NFR-UX-###) and to safety requirements SAF-MOU-001 to SAF-MOU-003 in the SRS, Safety-Critical Requirements.

Table 4: FMEA: sensory load, feedback, progress views and refinement

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-MOU-001 | Child-facing screens; audio feedback | Screens or feedback present more sound, colour, movement or flashing than the child can manage. | Decorative animation, repeated or loud sounds, bright colours or many controls; children react differently to sensory input (SRC-009.05; FACT-003); no adult may be watching (CA-001). | The child becomes overwhelmed or distressed and stops practising; rapid flashing can also trigger seizures in rare cases. Loss: well-being; physical safety; the practice benefit. | Psychological/ well-being; Physical | Eliminate: no flashing on any child-facing screen, and no animation while the child draws. Mitigate: let an adult turn audio off for each child, let long sounds be paused, and keep a calm style reviewed by adults including an educator or occupational therapist. Document/Test: inspect every child-facing screen against the calm-screen checklist. | SAF-MOU-003, NFR-UX-001, NFR-UX-006, NFR-UX-007, NFR-UX-010, NFR-UX-022 |
| H-MOU-002 | Feedback wording and style | Feedback tells the child that the drawing is wrong or bad, or uses error styling, after a missed part or a non-passing attempt. | Messages written around errors; warning colours, crosses or alarm sounds; wording not yet reviewed with educators or therapists (FACT-012). | The child feels they failed, loses confidence and avoids drawing (SRC-009.02). Loss: well-being; motivation to practise. | Psychological/ well-being | Eliminate: no error styling and none of the words “wrong”, “bad”, “fail” or “mistake” in any message to the child. Mitigate: acknowledge effort before showing what to change, and review wording with an educator or occupational therapist. Document/Test: check every approved message against the word list. | NFR-UX-004, NFR-UX-007, NFR-UX-015, NFR-UX-016, FR-LRN-022 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-MOU-003 | Feedback presentation | The child cannot tell what the feedback means or what to do next. | Feedback that needs reading, long messages, sound-only cues while audio is off, or a missed part shown only by colour to a child who does not see that colour. | The child repeats the same difficulty, gives up or needs an adult, and evaluable attempts are used up. Loss: the practice benefit; well-being. | Psychological/ well-being | Mitigate: show every message on screen and never by sound alone; keep messages short; use images instead of text; mark missed parts by more than colour. Document/Test: review all feedback with audio off and with colour hidden. | NFR-UX-002, NFR-UX-004, NFR-UX-015, NFR-UX-021 |
| H-MOU-004 | Feedback; guidance stages | The child comes to rely on Squigglr's help and does not practise drawing on their own. | Feedback completes the missed part for the child instead of pointing to it; prompts at every attempt; guidance that never fades. | The child does not move towards independent drawing, which is the aim of the practice (G.2). Loss: the practice benefit. | Psychological/ well-being | Eliminate: in guided lessons, never draw a missed part for the child. Mitigate: point to the missed part and let the child draw it; fade guidance automatically on a pass; keep the child's actual strokes. Document/Test: check that no feedback adds strokes to an attempt. | SAF-MOU-001, FR-LRN-014, FR-LRN-023, BR-004 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-MOU-005 | Progress summaries; attempt records | A progress view makes the child look more or less independent than they were. | Assisted attempts shown as independent; refined creative drawings mixed into lesson records; fewer attempts presented as better skill (SRC-005, Moustafa, 6:03). | An adult moves the child on too early or holds them back, or overrates their progress. Loss: suitable support decisions; the child's well-being. | Data integrity; Psychological/ well-being | Eliminate: never include creative-space drawings in lesson records or summaries. Mitigate: show assisted and independent attempts separately and never present the number of attempts as a measure of skill; use plain, non-clinical wording. Document/Test: check summaries of prepared histories with known assisted and independent attempts. | SAF-MOU-002, G.6, FR-LRN-012, BR-005, SAF-RAY-004, NFR-UX-017 |
| H-MOU-006 | Creative-space refinement | Refinement changes the drawing too much: points move further than intended, or parts are added or removed. | `REFINE_TOLERANCE` too large or not enforced; the refinement service returns a different drawing; the child cannot find how to restore the original (CA-001). | The child no longer sees the drawing as their own and is upset; work changed unexpectedly is a source of frustration (SRC-009.01; SRC-009.02). Loss: well-being; trust in the creative space. | Psychological/ well-being | Mitigate: limit every change to `REFINE_TOLERANCE`; let the child or the adult restore the original or turn refinement off; let the adult turn refinement off for each child. Document/Test: run the G.4 measurement on prepared adult-made drawings. | G.4, BR-005, NFR-UX-011 |

# 6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

# 7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

# 8 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

## 8.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

## 8.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

## 8.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

### Individual Disclosures

**Rayan:** I used Anthropic's Claude Opus 5.5 at medium effort, accessed through Claude Code, while preparing my sections of this document. I developed the ideas and made the project decisions in my sections myself, and Moustafa and I conducted the specialist interview and follow-up questionnaire. A major use of AI was organizing our sources: I used the model to sort through the interview transcript, questionnaire answers, and meeting notes, and to extract the specific statements, with their timestamps, that supported the ideas I wanted to include. This would have been very difficult to do by hand given the length of

the material. I also used it to turn my notes and decisions into draft wording, to improve sentence structure, clarity, and flow, and to help with LaTeX formatting. Through Claude Code, it also helped manage the repository (branches, commits, and merges) and helped me review my teammates' pull requests by checking their changes against our decisions and the rest of the document; I decided which review comments to post.

All ideas and decisions in my sections are my own. I reviewed every AI-assisted passage against my intended meaning and checked each cited statement against the original transcript, questionnaire, or meeting notes to confirm that it says what we claim it says.

**Ahmed:** I used OpenAI's ChatGPT (GPT-6) while writing my parts of the Hazard Analysis, particularly the H-AHM failure modes, their effects and mitigations, and the links to the SRS. All of the ideas behind my contributions are my own, and the project decisions were made through discussion with my teammates. I used AI to help put those ideas into words, improve grammar and sentence structure, and keep the technical descriptions clear and consistent. It also helped me express the analysis in the LaTeX table format and refine wording where my hazards overlapped with a teammate's work. Our team also used AI to help divide the workload fairly and evenly among members, taking into account the scope and complexity of each task. I personally used it to help write clear and descriptive pull request descriptions that accurately summarized the changes I made, making them easier for my teammates to review.

I reviewed the AI-assisted wording before including it, checked that the failure modes and mitigations reflected what I intended, and compared the requirement references with the SRS. The final choices about what to include and how to address each risk were mine and the team's, not the model's.

**Moustafa:** I used Anthropic's Claude Opus 5.5, accessed through Claude Code, alternating between medium and high effort depending on the task. Because the document was so large, I relied on it heavily to save time on project management. Connected to the GitHub CLI, it split my assignment into ten sub-issues so I could track my work in smaller pieces, checked which tasks I still had left, and made my commits and pull requests. For writing, I decided what each part should say and gave it my ideas, sometimes after brainstorming with it. It then wrote a first draft and the LaTeX for it, which I refined until it said exactly what I intended. In this document, that covered the H-MOU rows of the FMEA and their links to the SRS. I also used it for review: as teammates merged their sections, I had it read the document again to check that my additions stayed consistent with the rest and contradicted nothing, and it helped me review my teammates' pull requests. GitHub Copilot also gave a light automated review on a few of my pull requests.

All ideas in my sections are my own. I reviewed every AI-drafted passage against my intended meaning before including it, and I decided which flagged inconsistencies and review comments were valid and how to resolve them.

# 9 Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

   **Rayan:** Identifying the hazards was easier than expected because we already had good sources to work from. The transcript of our interview with three domain specialists on October 1, 2026 gave us concrete statements, with timestamps, about how the intended children use apps and what could go wrong for them. For example, the specialists said adults should decide progression rather than the app, that audio feedback should be possible to mute, and that Squigglr may be used at home, at school or in other settings. Each of these pointed directly to a failure mode we could analyse, so we were not guessing at hazards from scratch. Our meeting with our supervisor, Dr. Anand, on October 14, 2026 helped in the same way: once Dr. Anand explained that thresholds should be set by the adult for each child within a range, the hazards around wrong or extreme threshold values became clear. Because the SRS already recorded these sources, business rules and assumptions, we could trace each hazard and critical assumption back to a specific statement, which also made the mitigations easier to justify.

   **Ahmed:** What went well for me was having the learning activity requirements written in enough detail that I could actually think through how they might fail. I had been working on the guided-lesson flow in the SRS, so I already knew the steps from drawing a stroke to classifying an attempt, showing feedback and deciding whether the learner advances. Looking at each of those steps made it much easier to come up with specific hazards instead of just listing general problems that any website could have.

   It also helped that we were working on the SRS and Hazard Analysis at the same time. When I identified a failure mode, I could check whether an existing requirement already addressed it and point to that requirement directly. For duplicate stylus events and incorrect coordinate mapping, I could see that we needed two more explicit safeguards in the SRS. I

think that connection between the two documents was the strongest part of my work, because the hazards led to concrete requirements that we can eventually test.

**Moustafa:** What went well was how much this deliverable changed the way I think about what we're building. When I think of developing a drawing app for kids, the last thing that comes to mind is hazards and safety. But going through my rows made me realize how many gaps we could have had if we hadn't thought about this. For example, I never would have guessed a drawing app could cause physical harm, but rapid flashing can trigger seizures in people with photosensitive epilepsy, which is why WCAG limits flashing to three times a second. That became H-MOU-001 and SAF-MOU-003. I think thinking in terms of hazards early on is going to be really useful for my career in general, not just for this project.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   **Rayan:** The main pain point was that my hazards kept changing as other parts of the documents changed. I wrote my first set of hazards and critical assumptions based on the SRS as it stood at the time, where the team would set fixed values for thresholds such as `DIFFICULTY_STREAK` and `ATTEMPT_LIMIT`. In our meeting on October 14, 2026, Dr. Anand advised that suitable values differ from child to child, so I changed those thresholds to be set by the supporting adult within a range the team defines. When I reviewed the hazard analysis again, I found that this changed the hazards: the risk was no longer only a wrong team-chosen value, but also an adult choosing a value at the edge of the range, so I added a new hazard and safety requirement for it. The same thing happened when I added the improvement rule, which can extend a lesson while a child is improving, and when we confirmed that the child does not navigate inside a lesson. Each time, I had to go back and update the affected hazards, critical assumptions and mitigations so they matched the SRS. It was time-consuming, but it showed me how closely the hazard analysis depends on the requirements, and that it has to be reviewed again whenever the SRS changes.

   **Ahmed:** The hardest part was deciding where one hazard ended and another began. A mistake in the drawing input can lead to a wrong classification, which can then lead to incorrect feedback or an inappropriate stage change. At first it was tempting to describe all of those consequences in every row. I had to keep going back and asking what the actual failure mode was, what caused it, and which requirement would prevent or reduce it.

   There was also some overlap with the hazards being written by other team members. In particular, my initial feedback hazard was too close to the work on confusing or discouraging feedback. After reviewing that

feedback, I narrowed my row to a technical problem: the system choosing the wrong pre-approved message for the classification it produced. This left the wording, tone and sensory aspects to the other hazard owner. Another challenge was keeping the references aligned as the SRS changed. Whenever we clarified a lesson rule or added a safety requirement, I had to check the FMEA again rather than assuming the old wording still worked. It took extra revisions, but the result was more precise and easier to review.

**Moustafa:** The main pain point was that some of my hazards depended on requirements nobody had written yet. For example, the hazards about refinement changing a drawing too much and progress views misleading adults belong with Muhammad's creative-drawing and progress-summary requirements, which weren't written when I was doing my rows. I couldn't just leave the mitigation column empty, so where a goal or business rule already covered the hazard, I traced to that, and where nothing did, I wrote my own safety requirements, SAF-MOU-001 and SAF-MOU-002. It also meant my rows came in late, because I needed my own usability sections done first so the hazards had something to point to.

3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?

   **Rayan:** Before this deliverable, the team had already thought of two of my risks. Because Squigglr runs in a browser, we expected that attempts could be lost or not saved on an unreliable browser or device (H-RAY-002). We also knew that adults might read a child's records as a diagnosis (H-RAY-003), which is why the problem statement already avoids any diagnostic claims.

   The other five risks I only thought of while doing the hazard analysis. They came about because this was the first time I had to work out the requirements for the application in detail, while writing the SRS alongside this document. Most of them came from the critical assumptions, and several of those assumptions came from our interview with the specialists on October 1, 2026. The specialists said children can use the app through images and icons while adults handle setup, which led to the assumption that an adult sets up each session. Asking what happens if no adult is actually watching led to the risk of a child reaching adult-only actions (H-RAY-001), and, together with defining what happens after repeated failed attempts, to the risks of a child becoming distressed (H-RAY-004) and of a lesson being restarted straight after it ends (H-RAY-005). The specialists also said Squigglr may be used at home, at school or in other settings, which led to the assumption that a device may be shared and to the risk of a child's records being saved under the wrong profile (H-RAY-006). The last risk came from our meeting on October 14, 2026, where

Dr. Anand advised that adults should set the thresholds for each child; this raised the risk of a threshold being set to a value that does not suit the child (H-RAY-007).

Looking back, most of my risks only became visible once the requirements were concrete enough to question. The risks we knew about beforehand were general ones that apply to almost any browser app, while the ones I found during this deliverable were specific to how children and adults will actually use Squigglr. This showed me that the hazard analysis cannot be done well in isolation; it depends on knowing the requirements and on hearing from people who understand the users.

**Ahmed:** Before we started this deliverable, I had already thought about the possibility of drawing input being incomplete and of an incorrect attempt being allowed to pass. Those were concerns that came up naturally while defining how a guided lesson should evaluate strokes and move a learner to the next stage. They became H-AHM-001 and H-AHM-004 once I described their causes, effects and mitigations more systematically.

The FMEA made me consider several less obvious failures. For H-AHM-002, I started thinking about what happens when input events are reported more than once. Two overlapping strokes might be perfectly intentional, but one event being duplicated by the software is a different problem, and simply deleting repeated-looking lines would not solve it. H-AHM-003 came from looking at how the canvas changes size: even if the stylus reports valid points, a scaling or coordinate-mapping mistake could shift the drawing before it is scored. Those cases led to the new input-integrity safety requirements.

I also looked beyond the drawing itself. H-AHM-005 came from asking whether a correct classification could still produce the wrong feedback if the message lookup used the wrong stage or identifier. H-AHM-006 came from tracing the failure streak, improvement rule and adult-approval step together; an error there could suggest a stage change at the wrong time or apply one without permission. Finally, H-AHM-007 came from considering what happens when a learner's saved settings are changed or profiles are switched. That exercise showed me that a lot of meaningful risks are not obvious until you follow the data and decisions across the entire workflow.

**Moustafa:** Most of my risks were things we had already thought about before this deliverable, because we planned against them in the presentation we gave the specialists on October 1. We had already decided that feedback would highlight the unfinished part of a drawing instead of failing the attempt or finishing it for the child, that there would be no scores and fewer attempts wouldn't mean better skill, and that refinement would only tidy the child's own drawing, with the original always restorable. Those became H-MOU-002, H-MOU-004, H-MOU-005 and H-MOU-006. The other two came up while doing this deliverable. Sensory overload (H-MOU-001) came from the questionnaire, where one specialist

warned about excessive visual or auditory stimuli, and the flashing part came from WCAG while I was writing the accessibility requirements. Confusing feedback (H-MOU-003) came from the same work: once I let adults mute audio and said feedback can't rely on colour alone, I started asking what happens to a child who gets sound-only or colour-only feedback, or feedback they need to read.

4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

   **Rayan:** Most software carries risks that have nothing to do with physical harm. One is privacy and security. Almost every application stores some information about its users, and if that information is exposed, accessed by the wrong person or mixed up between users, it can harm those users and destroy their trust in the product. This matters even more when the users are vulnerable. In Squigglr, for example, the records describe a child's difficulties, so they must only be visible to the adults responsible for that child. A second is data integrity and availability. People make decisions based on what software shows them, so data that is lost, corrupted or shown incorrectly can lead them to the wrong decision, even when nothing visibly crashes. In our case, an adult choosing a child's next lesson relies on the attempt records being complete and accurate. A third is psychological harm. Software that frustrates, confuses or misleads its users can affect their well-being or lead them to wrong conclusions, for example by implying something about a person that the software cannot actually judge. For Squigglr, that means avoiding repeated failure without a break and making sure progress summaries are never read as a diagnosis. These risks are important to consider because they are easy to overlook when there is no obvious physical danger, yet they can cause real harm to users and to the people who rely on the software.

   **Ahmed:** One important type is data integrity risk. Software can appear to work while quietly saving or interpreting the wrong information. That matters for Squigglr because a drawing could be captured incorrectly, marked as passing when it should not, or saved under the wrong result. Adults rely on those records to understand what happened during practice, so an inaccurate record is not just a technical bug; it can influence their next decision.

   Another type is privacy and security risk. A system may expose one person's information to someone who should not have access to it, or mix records between different users. Squigglr may be used on shared devices, and its records concern children, so we need to think about who can see each learner's history and how the active learner is identified. Psychological or well-being risks also matter. Incorrect feedback, repeated failures or unexpected progression can confuse or frustrate a child even if there is no physical injury. These examples reminded me that software

safety is about more than preventing a crash or a hardware failure. It also means making sure people can trust what the system tells them and that it behaves in a way appropriate for its users.

**Moustafa:** Two other types of risk are accessibility risk and legal or compliance risk. Accessibility risk is when some people simply can't use the software, and it's easy to miss because teams usually test on people like themselves. For Squigglr, a child who can't read yet, is colour-blind or has trouble with fine motor control could fail to use the app without anyone noticing, which is the opposite of what we're building it for. That's what most of my usability and accessibility requirements are there to prevent. Legal and compliance risk is when software breaks a law or regulation, which can get it pulled or fined even if it works fine. One of our specialists pointed out that WCAG is adopted as law in many countries, and Squigglr keeps records about children, so we have to think about both accessibility and privacy rules before we can put it in front of real families or schools.